

Industrial Functional Programming

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1 Release handlings

Release

- A standalone node containing the erts and a copy of the VM +
- Set of applications (new or reused) +
- The depending OTP applications
- Configuration, bootfile and startscript

Release handling

- Standard configuration and makefile
- systools
- rebar - general build tool
- relx - release and target system builder
- reltool - graphical front-end based build customization
- etc.

Release structure

- lib - different versions of different applications
- erts - binaries for the Erlang runtime system
erts/bin - executables and shell scripts
(erlc, epmd, erl, escript, typer, etc)
- release - start_erl.data + rel, script and boot files for each release
- bin - links to the default erts-vsn/bin files

Minimal release

- kernel and stdlib
- usually include sasl (System Architecture Support Libraries)
 - needed for software upgrade
- start_sasl.rel

```
{release, {"Erlang/OTP","19"}, {erts, "8.3"},  
  [{kernel,"5.2"},  
   {stdlib,"3.3"},  
   {sasl, "3.0.3"}]}.
```

Chat release

```
{release, {"chat", "1.0"}, {erts, "8.2.2"},  
  [{kernel, "5.1.1"},  
   {stdlib, "3.2"},  
   {chat_app, "1.0"}]  
}.
```

.rel file

```
{release, {RelName, RelVsn}, {erts, ErsVsn},  
  [{AppName, AppVsn},  
   {AppName, AppVsn, Type},  
   {AppName, AppVsn, IncApps},  
   {AppName, AppVsn, Type, IncApps}]  
}.
```

```
Type: load | none | permanent (default)  
      | transient | temporary
```


Creating .script and .boot files

```
systools:make_script("chat",  
                    [{path, ["path_to_/ebin"]}])).
```

```
erl -boot chat -pa path_to_ebin/
```

Some options:

```
no_module_tests, src_tests, silent, {outdir, Dir}  
no_dot_erlang, local, warnings_as_errors
```

Creating a release package

```
cp chat.config sys.config
```

```
systools:make_tar("chat",  
                  [{erts, "/usr/lib/erlang/"},  
                   {path, ["path_to_/ebin"]}])).
```

Creating a release package

- create a bin directory and copy start, start_erl, to_erl, run_erl
- modify ROOTDIR in start_erl
- create start_erl.data in releases (erts version and release version)
- create log directory

Creating a release package

```
mkdir bin
cp erts-9.0/bin/start.src bin/start
cp erts-9.0/bin/to_erl bin/
mkdir log
```

```
ROOTDIR=path_to_/myrel/chat
```

```
bin/start
bin/to_erl /tmp/
```

USE CTRL-D !!!